HW2
313

(2) 1) **D5.** We have a feed that is a binary mixture of methanol and water (55.0 mol% methanol) that is sent to a system of two flash drums hooked together. The vapor from the first drum is cooled, which partially condenses the vapor, and then is fed to the second flash drum. Both drums operate at a pressure of 1.0 atm and are

(10)

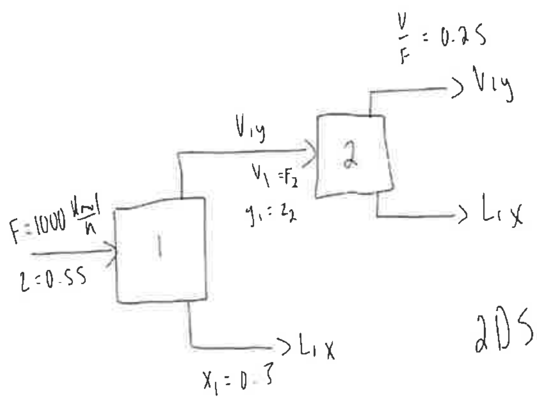
D10.

[redacted]

adiabatic. The feed rate to the first drum is 1000.0 kmol/h. We desire a liquid product from the first drum that is 30.0 mol% methanol ($x_1 = 0.300$). The second drum operates at a fraction vaporized of $(V/F)_2 = 0.250$. The equilibrium data are in Table 2-8 in Problem 2.D1.

a. Find the following for the first drum: y_1, T_1, $(V/F)_1$, and vapor flow rate V_1.

b. Find the following for the second drum: y_2, x_2, T_2, and vapor flow rate V_2.



2D5a) $F = L_1 + V_1 \rightarrow 1 = \frac{L}{F} + \frac{V}{F}$

$Fz = L_1 x_1 + V_1 y_1 \quad \frac{L}{F} = 1 - \frac{V}{F}$

$z = \frac{L_1 x_1}{F} + \frac{V_1 y_1}{F}$

table

$x_1 = 0.3 \quad y_1 = 0.665 \quad T_1 = 78°C$

$z = (1 - \frac{V_1}{F})x_1 + \frac{V_1}{F} y_1$

$0.55 = (1 - \frac{V}{F})(0.3) + \frac{V_1}{F}(0.665)$

$0.55 = 0.3 - 0.3\frac{V_1}{F} + 0.665\frac{V_1}{F}$

$0.25 = 0.365\frac{V_1}{F} \rightarrow \frac{V_1}{F} = 0.685$

$\frac{V_1}{F} = 0.685 \rightarrow V_1 = 0.685(1000) = 685 \ \frac{kmol}{h}$

2D5b) $V_1 = F_2 = 685 \ \frac{mol}{h}$

$y_1 = z_2 = 0.665$

$Fz_2 = L_2 x_2 + V_2 y_2$

$1 = \frac{L}{F} + \frac{V}{F} \rightarrow \frac{L}{F} = 1 - \frac{V}{F}$

$z_2 = (1 - \frac{V_2}{F})x_2 + \frac{V_2}{F} y_2$

$0.665 = (1 - 0.25)x_2 + 0.25 y_2$

$x = 0.6 \quad y = 0.825 \quad T = 71.2$

$x = 0.7 \quad y = 0.87 \quad T = 69.3$

$y = 0.825 + \frac{(x-0.6)(0.87-0.825)}{0.7-0.6}$

$y = 0.825 + 0.45(x - 0.6)$

$y = 0.825 + 0.45x - 0.27$

$y_2 = 0.45x_2 + 0.555$ (sub in)

$0.665 = 0.75x_2 + 0.25(0.45x_2 + 0.555)$

$0.665 = 0.75x_2 + 0.1125x_2 + 0.13875$

$0.52625 = 0.8625x_2 \rightarrow x_2 = 0.61$

$y_2 = 0.45(0.61) + 0.555 = 0.83$

$T = 71.2 + \frac{(x-0.6)(69.3-71.2)}{0.7-0.6} = 71.2 - 19(x - 0.6)$

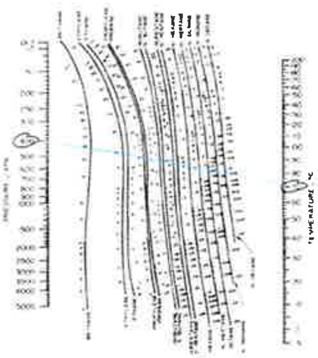
$T_2 = 71.2 - 19(0.61 - 0.6) = 71°C$

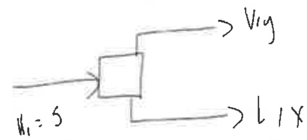
$\frac{V_2}{F} = 0.25 \rightarrow V_2 = 0.25(685) = 171.25 \ \frac{kmol}{h}$

R

* Σ

2) D10. We have a mixture that is 35.0 mol% n-butane with unknown amounts of propane and n-hexane. We are able to operate a flash drum at 400.0 kPa and 70.0°C with $x_{C6} = 0.700$. Find the mole fraction of n-hexane in the feed, z_{C6}, and the value of V/F.





$C_3 \; z_1 = \quad K_1 = 5$

$C_4 \; z_2 = 0.35 \quad K_2 = 2$

$C_6 \; z_3 = \quad K_3 = 0.3$

$x_{C6} = 0.7$

$x_i = \frac{z_i}{1+(K_i-1)V/F} \qquad y_i = K_i x_i$

$x_{C4} = \frac{0.35}{1+(2-1)V/F} = \frac{0.35}{1+1\,V/F}$

$x_{C4} + x_{C6} + x_{C3} = 1$ (with $x_{C6} = 0.7$)

$x_{C3} = 0.3 - x_{C4}$

$x_{C6} = \frac{z_3}{1+(0.3-1)V/F} = 0.700$

$z_1 + z_2 + z_3 = 1$ (with $z_2 = 0.35$)

$z_1 + z_3 = 0.65$

$z_1 = 0.65 - z_3$

$x_{C3} = \frac{z_1}{1+(5-1)V/F} = \frac{z_1}{1+4V/F}$

R.R.

$z_3 = 0.7(1-0.7V/F) = 0.7 - 0.49V/F$

$z_1 = x_{C3}(1+4V/F) \rightarrow 0.65 - 0.7(1-0.7V/F) = -0.05 + 0.49V/F$

$0.35 = x_{C4}(1+V/F)$

$\sum \frac{(K_i-1)z_i}{1+(K_i-1)V/F}$

$$\frac{4(-0.05+0.49V/F)}{1+4V/F} + \frac{0.35}{1+V/F} + \frac{-0.7[0.7\cancel{(1-0.7V/F)}]}{\cancel{1-0.7V/F}} = 0$$

$$\frac{-0.2+1.96V/F}{1+4V/F} + \frac{0.35}{1+V/F} - 0.49 = 0$$

$\frac{V}{F} = 0.46$

$z_3/z_{C6} = 0.7 - 0.49(0.46)$

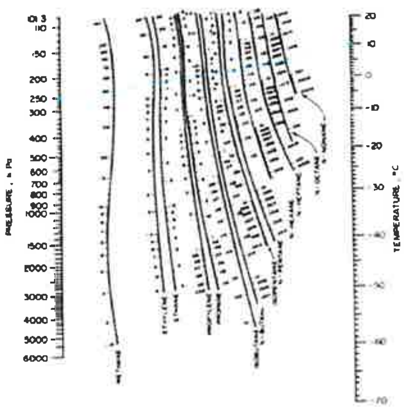
$z_{C6} = 0.47$

$(V/F)_1 = 0.3 \rightarrow f(0.3) = -0.044$

$(V/F)_2 = 0.4 \rightarrow f(0.4) = -0.015$

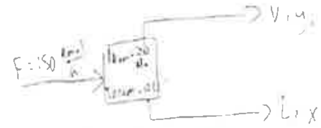
$(V/F)_3 = 0.45 \rightarrow f(0.45) = -0.005 = 0.005$ x

$(V/F)_4 = 0.46 \rightarrow f(0.46) = -0.003 < 0.005$



3) ②

16.* A feed that is 50.0 mol% methane, 10.0 mol% n-butane, 15.0 mol% n-pentane, and 25.0 mol% n-hexane is flash distilled. F = 150.0 kmol/h, p_{drum} = 250.0 kPa, and T_{drum} = 10.0°C. Use DePriester charts or Eq. (2-28). Find V/F, x_i, y_i, V, and L.



$C_1, z_1 = 0.5$ $K_1 = 80$
$C_4, z_2 = 0.1$ $K_2 = 0.57$
$C_5, z_3 = 0.15$ $K_3 = 0.16$
$C_6, z_4 = 0.25$ $K_4 = 0.06$

NR

$$f\left[\frac{V}{F}\right] = \sum_{i=1}^{4} \frac{(K_i - 1)z_i}{1+(K_i-1)V/F}$$

$$= \frac{(80-1)(0.5)}{1+(80-1)V/F} + \frac{(0.57-1)0.1}{1+(0.57-1)V/F} + \frac{(0.16-1)0.15}{1+(0.16-1)V/F} + \frac{(0.06-1)0.25}{1+(0.06-1)V/F}$$

$$f\left(\frac{V}{F}\right) = \frac{39.5}{1+79 V/F} + \frac{-0.043}{1-0.43 V/F} + \frac{-0.126}{1-0.84 V/F} + \frac{-0.235}{1-0.94 V/F}$$

$$\frac{df_A}{d(V/F)} = -\sum \frac{(K_i-1)^2 z_i}{[1+(K_i-1)V/F]^2} = -\left[\frac{(80-1)^2 0.5}{[1+79 V/F]^2} + \frac{(0.57-1)^2 0.1}{[1-0.43 V/F]^2} + \frac{(0.16-1)^2 0.15}{[1-0.84 V/F]^2} + \frac{(0.06-1)^2 0.25}{[1-0.94 V/F]^2}\right]$$

$$= -\left[\frac{3120.5}{(1+79 V/F)^2} + \frac{0.01849}{(1-0.43 V/F)^2} + \frac{0.10584}{(1-0.84 V/F)^2} + \frac{0.2209}{(1-0.94 V/F)^2}\right]$$

① $(V/F)_1 = 0.3$ $\frac{df_A}{d(V/F)} = -5.76$ $f(0.3) = 1.05$

$$\left[\frac{V}{F}\right]_2 = \left[\frac{V}{F}\right]_1 - \frac{f_A}{[df_A/d(V/F)]} = 0.3 + \frac{1.05}{5.76} = 0.48$$

② $(V/F)_2 = 0.48$ $\frac{df_A}{d(V/F)} = -3.12$ $f(0.48) = 0.32 > 0.005$

$$\left(\frac{V}{F}\right)_3 = 0.48 + \frac{0.32}{3.12} = 0.58$$

③ $(V/F)_3 = 0.58$ $\frac{df_A}{d(V/F)} = -2.93$ $f(0.58) = 0.024$

$$\left[\frac{V}{F}\right]_4 = 0.58 + \frac{0.024}{2.93} = 0.588$$

④ $(V/F)_4 = 0.588$ $\frac{df_A}{d(V/F)}$ 2.916 $f(0.588) = 4.78 \times 10^{-4} < 0.005$

$\frac{V}{F} = 0.588$

$$x_i = \frac{z_i}{1+(K_i-1)V/F} \qquad y_i = K_i x_i$$

C_1 $x_1 = 0.011$ $y_1 = 0.88$
C_4 $x_2 = 0.134$ $y_2 = 0.08$
C_5 $x_3 = 0.296$ $y_3 = 0.05$
C_6 $x_4 = 0.557$ $y_4 = 0.07$

$$\boxed{\begin{array}{ll} C_1, z_1 = 0.5 & K_1 = 80 \\ C_4, z_2 = 0.1 & K_2 = 0.57 \\ C_5, z_3 = 0.15 & K_3 = 0.16 \\ C_6, z_4 = 0.25 & K_4 = 0.06 \end{array}}$$

F = 150 kmol/h $\frac{V}{F} = 0.588$ $V = 0.588(150) = 88.2$ kmol/h

$L = F - V = 150 - 88.2 = 61.8$ kmol/h

4)

D24. We plan to separate a mixture of propane and n-hexane at 300 kPa.

a. Using the data in the DePriester charts, plot $y_{propane}$ vs. $x_{propane}$ and T vs $x_{propane}$ for this mixture at 300 kPa.

b. If the feed is 30.0 mol% propane, and V/F = 0.40, what are the liquid and vapor mole fractions and the drum temperature? Solve graphically.

c. If $y_{propane}$ = 0.80 and the feed is $z_{propane}$ = 0.60, what is the value of V/F?

d. Use the Rachford-Rice equation to check the answers obtained in parts b and c.

b) (0.09, 0.62)

T = 73°C

c) y = 0.8

$\frac{V}{F} = 0.68$

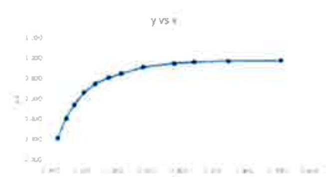
d)

$-0.0045 < 0.005$

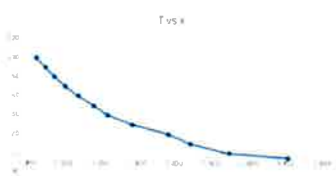
$0.0027 < 0.005$

a)

T	K3	K6	x3 propane liquid	y3 propane vapor	x6 nhexane l	y6 nhexane v
-5	1.4	0.023	0.710	0.990	0.290	0.007
0	1.8	0.03	0.548	[redacted]	0.452	0.014
10	2.2	0.044	0.443	0.976	0.557	0.024
20	2.5	0.067	0.383	0.959	0.617	0.041
30	3.2	0.109	0.288	0.922	0.712	0.078
40	3.9	0.18	0.220	0.860	0.780	0.140
50	4.5	0.22	0.182	0.820	0.818	0.180
60	5.4	0.28	0.141	0.759	0.859	0.241
70	6.4	0.37	0.104	0.669	0.898	0.331
80	7.3	0.49	0.075	0.547	0.925	0.453
90	8.2	0.62	0.050	0.411	0.950	0.589
100	9	0.8	0.024	0.220	0.976	0.780

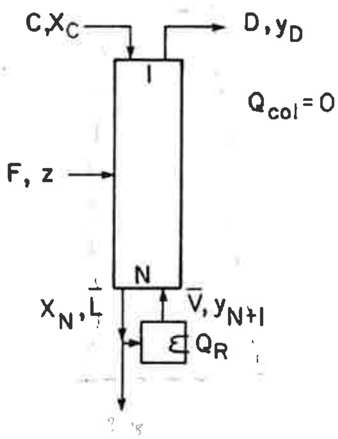


T	x3
-5	0.710
0	0.548
10	0.443
20	0.383
30	0.288
40	0.220
50	0.182
60	0.141
70	0.104
80	0.075
90	0.050
100	0.024



5)

D3.* A distillation column separating ethanol from water is shown in the following figure. Pressure is 1 kg/cm². Instead of having a condenser, a stream of pure, saturated liquid ethanol is added directly to the column to serve as the reflux. The feed to the column is 40 wt % ethanol at −20°C. Distillate concentration is 80 wt % ethanol, and bottoms composition is 5 wt % ethanol. A total reboiler is used, and the boilup is a saturated vapor. The reflux stream is input at C = 1000 kg/h. Find the external boilup rate, $\bar{V}$. Note: Set up the equations, solve in equation form for $\bar{V}$, including explicit equations for all required terms, read off all required enthalpies from the enthalpy composition diagram (Figure 2-4), and then calculate a numerical answer.



V = 1174.4 kg/h

Homework 2

1)

a)

b)

a) $x_1 = 0.3$

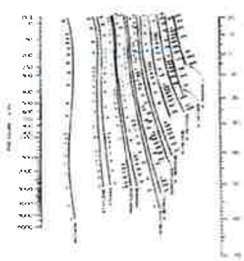
$y_1 = 0.665$

$T_1 = 78\,^\circ C$

$V_1 = 685 \frac{kmol}{h}$

$\frac{V}{F}_1 = 0.685$

b)



2)

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3)

D16.* A feed that is 50.0 mol% methane, 10.0 mol% n-butane, 15.0 mol% n-pentane, and 25.0 mol% n-hexane is flash distilled. F = 150.0 kmol/h, $p_{drum} = 250.0$ kPa, and $T_{drum} = 10.0\,^\circ C$. Use DePriester charts or Eq. (2-28). Find V/F, x_i, y_i, V and L.